

gies like multiple-input multiple-output (MIMO), VoLTE, LTE-Advanced and LTE Carrier Aggregation to improve customer experience through enhanced spectrum efficiency and resulting high data speeds. 4G roll-out and 5G-related R&D in telecom will create opportunities for the T&M industry.

Compared to previous generations of mobile networks, 5G presents a number of new design and test challenges. While the technologies that will constitute 5G are still being worked on, the ITU-R has identified three main usage scenarios for 5G: enhanced mobile broadband with up to 20Gbps peak data rate, massive machine-type communications, and ultra-reliable and low-latency communications.

To meet the requirements for this next generation of mobile communications, researchers are looking at a wide range of enabling technologies in RF, microwave and millimetre-wave frequencies—ranging from advanced waveforms and multiple-access schemes to multiple-antenna techniques such as massive MIMO and beamforming. Being the first and the best in 5G depends on tools that let operators explore new signals, scenarios and topologies.

Automation is the new technology trend in power sector and hence the demand for automated products is growing. Accordingly, new technological advancements like data-logging digital multimeters and power clamp meters with automatic calculation will experience good growth. Online transmission-line monitoring systems, compact

STRENGTHS

- Continuous technological advances in testing equipment, such as multi-functional testing kits
- The ongoing practice of procuring the most reliable testing equipment
- Adoption of the best testing practices by OEMs
- Stringent quality and safety standards for end products

WEAKNESSES

- Enforcement of quality standards pertaining to harmonics level, etc not yet mandated by the government
- Need to revive important quality norms like ISI

SWOT

OPPORTUNITIES

- Introduction of various government initiatives like Make in India and Digital India
- Growth of domestic manufacturing in areas like defence and telecommunications
- Growth in power demand leading to the requirement of reliable, accurate and efficient equipment
- Requirement for efficient power generation, transmission and distribution
- Manufacturers' increasing awareness about predictive maintenance
- Proliferation of smart devices

THREATS

- Lack of industry awareness about the usability and functionality of testing equipment
- Various unscrupulous players not adhering to standard engineering and commercial practices
- Quality affected because the government's tender business is done only on L1 basis
- Increasing focus on the quality for end-user segments is time-consuming, affecting the growth of T&M industry

SWOT analysis of T&M market

Tan-delta testing machines, automatic breaker testing machines and discrete monitoring of power loss at each feeder by discoms are some of the application areas.

As regards user experience, touch-screen technologies are going to revolutionise the T&M environment. A combination of the right hardware with a 'designed for touch' user interface (UI) will enable engineers to use equipment easily and efficiently. Designed-for-touch UIs are crucial for creating an effective and usable touch-screen environment; it is not enough to simply add a touch-screen to an existing UI. T&M equipment are designed to identify, isolate and ana-

lyse a system under test as quickly and efficiently as possible. Providing an effective touch-screen on the instrument can reduce the time it takes to configure the system and to analyse signals.

The way forward

Survey respondents feel that it is very important to make customers aware and convinced about quality, safety and maintenance practices through the use of right T&M instruments. Regulatory bodies can help to spread this awareness.

Moreover, T&M companies need to work as solution partners with decision makers in the electronic system design and manufacturing (ESDM) space by offering leading-edge electronics design and test services. By adopting the latest and the best T&M solutions, ESDM players can make the elusive breakthroughs that advance technology. That will lead to ideas, systems and products that could make the world a better place and the technology business more productive. **EFY**

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